

ColdTrack AI: Evaluation Artifact (Iteration 1)

AIC COMPFEST 18 Final · Smart Logistics · results tag: iter1 · compared with the baseline artifact

What is this document?

ColdTrack AI watches the sensors of a refrigerated truck and warns the operator before the cargo gets too warm. The baseline artifact measured the prototype from the qualifying round. This document measures the rebuilt system with the same test trips, so the two can be compared.

The rebuilt system has three parts. A physics engine estimates the cargo temperature from the cabin sensor, counts down to a breach, and sets the status (AMAN = safe, WASPADA = warning, KRITIS = critical). A set of rules checks whether the temperature sensor itself is faulty. A learned model (GRU) names three events (door left open for 20 minutes or more, sudden outside heat, faulty sensor) and forecasts the cargo temperature. The countdown no longer depends on the fault classifier, which was the main failure of the baseline.

Test data: 106 simulated trips (16,336 one-minute snapshots, each with the trip history up to that minute), the same held-out trips as the baseline. The rebuilt model was trained on a new physics simulator, not on these trips, so for it this is data from a different generator. The baseline was trained on data from the same generator as the test trips.

Summary

The rebuilt system passed 7 of 19 checks (the baseline passed 8 of 23 older checks; the lists differ because the outputs changed). The most dangerous baseline failure is fixed: when a breach was less than 30 minutes away, the baseline showed green in 32% of cases; the rebuilt system shows green in 0%. On these test trips, the rebuilt system still fails in other ways:

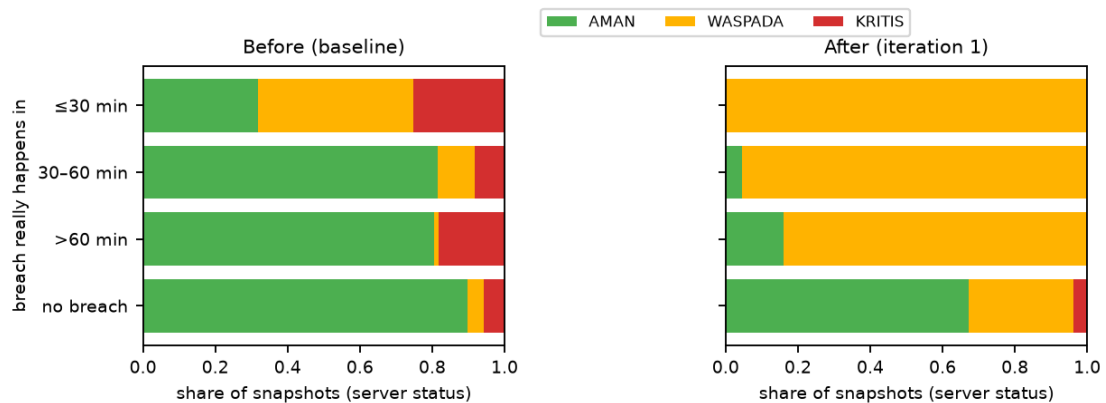
#	What we found	Why it matters	Severity
N1	Urgent cases are never shown as green, but they are never shown as critical either: 100% warning, 0% critical. The countdown is shown in 0% of urgent cases (baseline 62%).	The operator is warned but not told how much time is left.	Critical
N2	Healthy trucks get a warning in 33% of snapshots (baseline 6%).	Too many false warnings teach operators to ignore them.	High
N3	The temperature forecast is off by 1.41 °C on average (baseline 0.20 °C). Guessing “no change” gives 0.19 °C.	The forecast chart would mislead the operator.	High
N4	Door open and sudden outside heat are rarely named: 0% and 15% (baseline 59% and 44%).	The operator is not told what to fix.	High
N5	Small sensor noise (0.1 °C) drops correct fault names from 26% to 5%.	Real sensors are noisier than the simulator.	High
N6	A sensor that freezes 30 minutes before a real breach is not flagged (0%, baseline 10%), but the status is green in only 0% of those cases (baseline 73% judged healthy).	A stuck sensor still hides the fault type.	Medium
N7	The test trips come from the old generator and have no cargo mass; urgent cases exist only for dairy and vegetables/fruit (plus one vaccine snapshot).	Several results above may reflect the gap between the two simulators, not the truck.	Medium

1. Before and after

Same test trips and, where the measurement still exists, the same pass mark. Server numbers come from real requests to the API; the rebuilt system was sent 1,634 requests (every 10th snapshot), the baseline 8,122.

What we measure	Before (baseline)	After (iteration 1)	Goal
Checks passed	8 of 23	7 of 19	lists differ
Urgent cases (breach within 30 min) shown as green	32%	0%	5% or less
Urgent cases shown as critical	25%	0%	no goal set
Urgent cases with a countdown	62%	0%	80% or more
Healthy snapshots with a warning or critical status	6%	33%	10% or less
Door open named correctly	59%	0%	50% or more
Sudden outside heat named correctly	44%	15%	50% or more
Faulty sensor named correctly	67%	62%	50% or more
Forecast error at 30 min	0.20 °C	1.41 °C	below 0.8 °C
Urgent cases treated as safe with 0.3 °C sensor noise	82%	50%	50% or less
Sensor frozen during a real breach: flagged as sensor fault	10%	0%	50% or more
Server response time (95th percentile)	15 ms	112 ms	below 300 ms

Before, "urgent treated as safe" meant the classifier said healthy, which hid the countdown. After, it means the status is AMAN. Fault names before came from 7 classes; after, from 3 events. The after-forecast predicts the cargo temperature, the baseline predicted the sensor reading; in the test trips the two are the same value.



Each bar groups snapshots by how soon the cargo really leaves its safe range. The top bars should be red. Before, a third of the top bar was green; after, it is almost all amber.

2. What we found, with evidence

N1 · Warning without a countdown

In the 632 urgent snapshots, the physics countdown was above its 60-minute display limit in 100% of cases, so nothing was shown. Its error on these cases is 235 minutes. The countdown projects a breach only when the cabin air is already above the limit. In the test trips the urgent cases are slow drifts: the sensor sits at the limit (median 4.0 °C for 2 to 4 °C cargo) and creeps upward, so the projection says the cargo never crosses. The warning that does appear comes from a second rule (cargo within 1 °C of the limit). The learned countdown (XGBoost) does no better here: error 230 minutes.

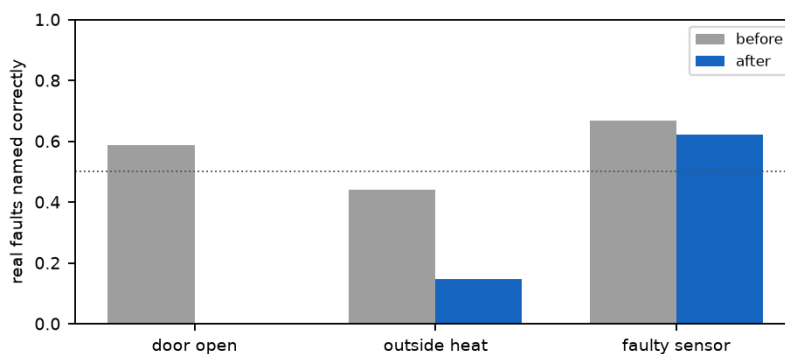
N2 · False warnings on healthy trucks

Every warning on a healthy truck comes from the same rule: estimated cargo within 1 °C of the upper limit. Dairy and vegetables/fruit allow only 2 to 4 °C, and in the test trips healthy cargo runs around 3.3 °C, inside that last degree. The new simulator keeps the same cargo at its 2.8 °C setpoint, and in the developers' 600-trip test on that simulator healthy trips got no warnings at all (ml/CHECKPOINT2_CONTEXT.md).

N3 · Forecast pulled toward the new simulator's settings

Average forecast error by cargo: vaccine 2.61 °C, frozen meat 2.05 °C, fish 0.49 °C, dairy 0.78 °C, vegetables/fruit 0.67 °C. The forecast leans toward the cooling setpoint of the new simulator (for example it predicts frozen meat about 2 °C warmer than it is), so it is worse than guessing "no change" (0.19 °C).

N4 · Door and outside heat not recognised

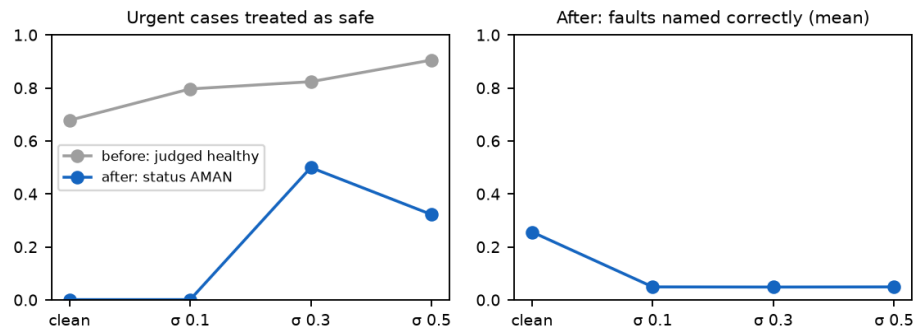


Door: even while the door is actually open, the model gives the door event a median probability of 0.04. In the test trips an open door barely changes the sensor reading; in the new simulator the cabin reading jumps 14 to 36 °C per minute, a known flaw of that simulator (no heat stored in walls and racks). The physics engine still raises the status to critical in 20 of the 58 snapshots with the door open.

Outside heat: the test trips model it as the truck standing still in the sun while the outside air does not warm (trip 43: 27.9 °C during the event against 30.1 °C before). The rebuilt model has no sunlight input and defines the event as the outside air rising to the hottest BMKG day, so the two definitions do not match.

Faulty sensor: 62% are named, all by the rules, with no false sensor alarms on other snapshots. The learned sensor score is weak (ranking quality 0.40, where 0.5 is chance), which is why the server uses the rules for this event.

N5 · Noise



With 0.3 °C noise, urgent cases shown as green rise from 0% to 50% (baseline: 82% judged healthy). At 0.5 °C the rate is 32%; we have not yet found why it is lower than at 0.3 °C. Fault naming collapses already at 0.1 °C. A 7-hour clock shift changes almost nothing (fault naming 25% against 26%).

N6 · Stuck sensor during a real breach

We froze the sensor for the last 30 minutes of 211 urgent snapshots. None were flagged as a sensor fault: the flat-reading rule needs 15 minutes of identical readings plus a sign that the temperature should have moved (door opened, outside air rising 3 °C), or 60 minutes without one. The status still stayed amber in 100% of these cases.

Checked and fine

The server returns exactly what the batch scoring computes (status agreement 100%, forecast difference 0.00 °C), with no errors in 1,634 requests. Response time is 90 ms typical and 112 ms at the 95th percentile, slower than the baseline because every request now carries the trip history.

3. Causes and plan

Causes marked (measured) were confirmed in this evaluation; the others are hypotheses to test with this suite before changing anything.

Finding	Cause	Planned change and how we will judge it
N1 no countdown	Countdown projects only when the cabin is above the limit; slow drifts never qualify (measured).	Add the recent sensor trend to the projection, or fall back to the learned countdown when physics cannot project. Judge: countdown shown in 80% of urgent cases.
N2 false warnings	1 °C margin rule on 2 °C wide cargo ranges (measured).	Scale the margin to the width of the cargo range. Judge: healthy warnings 10% or less while urgent cases stay 5% or less green.
N3 forecast	Forecast learned the new simulator's cooling setpoints (measured bias per cargo); cause of the bias itself is a hypothesis.	Include the observed recent temperature level more strongly (for example predict the change, not the level). Judge: error below 0.8 °C and better than "no change".
N4 door / heat	New simulator: door jumps too fast, no sunlight (measured difference with test trips).	Add wall and rack heat storage and a sunlight term to the simulator, retrain. Judge: door and heat named in 50% or more.
N5 noise	Training noise 0.073 °C from lab sensors (known); event head sensitive to it (measured).	Train with noise from 0.05 to 0.5 °C. Judge: fault naming drops 0.05 or less at 0.3 °C.
N6 stuck sensor	Flat-reading rule needs context or 60 minutes (measured).	Flag a flat reading after 15 minutes when the estimated cargo should be warming. Judge: 50% flagged, 20% or less green.
N7 test validity	Test trips from a different generator; no mass; 2 cargo types with urgent cases (measured).	Build a second test set from the new simulator with real product and mass mixes, and keep this one as the out-of-distribution check.

Appendix A: technical detail of the test groups

System under test: backend/app/inference.py (core engine + coldtrack_gru.onnx + coldtrack_ttb.onnx) as served by /api/v1/analyze, on the held-out test split of dataset v4, split by trip, with the same window selection as the baseline.
Reproduce: python -m eval.run_eval --tag iter1; pytest eval/test_baseline.py.

Group	Test cases	Metric(s)	What it does not capture
G1 Forecast	every snapshot; compared with “no change” and 5-minute trend	error at 15/30/60 min, worst case, error where temperature moves	Most snapshots are calm, hence the “no change” comparison.
G2 Events	v4 modes mapped A1 → door, A7 → heat, A5/A6 → sensor; A2/A3/A4/A8 have no event	share named correctly, precision, ranking quality of the learned score	Uncovered faults are judged only through status and countdown.
G3 Urgent breach	snapshots with a true breach within 10 / 30 min	countdown shown, status mix, why the countdown was hidden	Truth is the old simulator's breach time.
G4 Robustness	sensor noise 0.1/0.3/0.5 °C on whole trips; +7 h clock shift; sensor frozen for 30 min in urgent snapshots (every 3rd)	fault naming, urgent cases shown green, healthy warnings	Perturbations are synthetic; they show sensitivity, not field error.
G5 Server	every 10th snapshot sent to /api/v1/analyze with the trip history	status by breach bucket, countdown shown, latency, match with G1 to G3	Runs in-process (TestClient), so network time is excluded.

Appendix B: acceptance checks (iteration 1)

Result	Check
PASS	test_recall_per_event_at_least_0_5[sensor]
PASS	test_imminent_breach_not_shown_as_safe
PASS	test_sensor_noise_0_3C_does_not_hide_imminent_breach
PASS	test_timezone_shift_does_not_change_event_recall_much
PASS	test_imminent_breach_never_green_more_than_5pct
PASS	test_api_no_http_errors_and_latency_p95_under_300ms
PASS	test_api_matches_batch_scoring
FAIL	test_forecast_mae_t30_under_0_8C
FAIL	test_forecast_worst_case_error_bounded
FAIL	test_recall_per_event_at_least_0_5[door]
FAIL	test_recall_per_event_at_least_0_5[shock]
FAIL	test_healthy_windows_not_labelled_as_fault
FAIL	test_imminent_breach_ttb_shown_at_least_80pct
FAIL	test_test_split_covers_every_cargo_profile_for_imminent_breach
FAIL	test_sensor_noise_0_3C_event_recall_drop_at_most_0_05
FAIL	test_stuck_sensor_during_breach_is_flagged_not_safe
FAIL	test_imminent_breach_ttb_shown_by_api_at_least_80pct
FAIL	test_healthy_windows_not_over_alerted
FAIL	test_every_fault_mode_has_at_least_5_test_trips

Code commit: 648ba01 (plus uncommitted eval/ changes at generation time). Dataset: data/processed/v4_seed1000_700trips.parquet, split=test. Cargo mass: profile default (1,000 kg). Seeds fixed at 42.